

Oral gepotidacin versus nitrofurantoin in patients with uncomplicated urinary tract infection (EAGLE-2 and EAGLE-3): two randomised, controlled, double-blind, double-dummy, phase 3, non-inferiority trials.

PMID: 38342126 | Indexed in PubMed

Gepotidacin is a novel, bactericidal, first-in-class triazaacenaphthylene antibiotic that inhibits bacterial DNA replication by a distinct mechanism of action and a unique binding site, providing well balanced inhibition of two type II topoisomerase enzymes. Oral gepotidacin is under investigation to treat uncomplicated urinary tract infections. We aimed to compare the efficacy and safety of oral gepotidacin with that of nitrofurantoin in adolescent and adult female individuals with uncomplicated urinary tract infections. EAGLE-2 and EAGLE-3 were phase 3, randomised, multicentre, double-blind, double-dummy, non-inferiority (10% margin) trials, in which patients were enrolled at 219 centres worldwide. Patients assigned female at birth, non-pregnant, aged 12 years or older, weighing 40 kg or more, with two or more symptoms of dysuria, frequency, urgency, or lower abdominal pain, and with evidence of urinary nitrite, pyuria, or both were eligible for inclusion. Patients were randomly assigned (1:1) centrally by interactive response technology to receive oral gepotidacin (1500 mg twice daily for 5 days) or oral nitrofurantoin (100 mg twice daily for 5 days), with randomisation stratified by age category and history of recurrent uncomplicated urinary tract infections. Patients, investigators, and the sponsor study team were masked to treatment assignment. The primary endpoint, therapeutic response (success or failure) at test-of-cure (ie, day 10-13), was evaluated in randomly assigned patients with nitrofurantoin-susceptible qualifying uropathogens ($\geq 10^5$ colony-forming units [CFU] per mL) and who received at least one dose of study treatment. Conforming to regulatory guidance, therapeutic success was defined as combined clinical success (ie, complete symptom resolution) and microbiological success (ie, reduction of qualifying uropathogens to $< 10^3$ CFU/mL) without other systemic antimicrobial use. Safety analyses included patients who were randomly assigned and who received at least one dose of study treatment. The trials are registered with ClinicalTrials.gov, NCT04020341 (EAGLE-2) and NCT04187144 (EAGLE-3), and are completed. Studies were undertaken from Oct 17, 2019, to Nov 30, 2022 (EAGLE-2), and from April 23, 2020, to Dec 1, 2022 (EAGLE-3). 1680 patients in EAGLE-2 and 1731 patients in EAGLE-3 were screened for eligibility, of whom 1531 and 1605 were randomly assigned, respectively (767 in the gepotidacin group and 764 in the nitrofurantoin group in EAGLE-2, and 805 in the gepotidacin group and 800 in the nitrofurantoin group in EAGLE-3). After an interim analysis, which was prospectively agreed as a protocol amendment, both studies were stopped for efficacy. Thus, the primary analysis population included only patients who, at the time of the interim analysis data cutoff, had the opportunity to reach the test-of-cure visit or were known to not have attained therapeutic success before the test-of-cure visit. In EAGLE-2, 162 (50.6%) of 320 patients assigned gepotidacin and 135 (47.0%) of 287 patients assigned nitrofurantoin had therapeutic success (adjusted difference 4.3%, 95% CI -3.6 to 12.1). In EAGLE-3, 162 (58.5%) of 277 patients assigned gepotidacin and 115 (43.6%) of 264 patients assigned nitrofurantoin had therapeutic success (adjusted difference 14.6%, 95% CI 6.4 to 22.8). Gepotidacin was non-inferior to nitrofurantoin in both studies and superior to nitrofurantoin in EAGLE-3. The most common adverse event with gepotidacin was diarrhoea (observed in 111 [14%] of 766 patients in EAGLE-2 and in 147 [18%] of 804 patients in EAGLE-3), whereas the most common adverse event with nitrofurantoin was nausea (in 29 [4%] of 760 patients in EAGLE-2 and in 35 [4%] of 798 patients in EAGLE-3). Cases were mostly mild or moderate. No life-threatening or fatal events occurred. Gepotidacin is an efficacious oral antibiotic with acceptable safety and tolerability profiles. As a first-in-class investigational oral antibiotic with activity against common uropathogens, including clinically important drug-resistant phenotypes, gepotidacin has the potential to offer substantial benefit to patients. GSK and the US Office of the Assistant Secretary for Preparedness and Response, Biomedical Advanced Research and Development Authority.

In silico gepotidacin target mining among 33213 global *Neisseria gonorrhoeae* genomes from 1928 to 2023 combined with gepotidacin MIC testing of 22 gonococcal isolates with different GyrA and ParC substitutions.

PMID: 39004438 | Indexed in PubMed

The novel dual-target triazaacenaphthylene, gepotidacin, recently showed promising results in its Phase III randomized controlled trial for the treatment of gonorrhoea. We investigated alterations in the gepotidacin GyrA and ParC targets in gonococci by in silico mining of publicly available global genomes (n=33213) and determined gepotidacin MICs in isolates with GyrA A92 alterations combined with other GyrA and/or ParC alterations. We examined gonococcal gyrA and parC alleles available at the European Nucleotide Archive. MICs were determined using the agar dilution method (gepotidacin) or Etest (four antimicrobials). Models of DNA gyrase and topoisomerase IV were obtained from AlphaFold and used to model gepotidacin in the binding site. GyrA A92 alterations were identified in 0.24% of genomes: GyrA A92P/S/V+S91F+D95Y/A/N (0.208%), A92P+S91F (0.024%) and A92P (0.003%), but no A92T (previously associated with gepotidacin resistance) was found. ParC D86 alterations were found in 10.6% of genomes: ParC D86N/G (10.5%), D86N+S87I (0.051%), D86N+S88P (0.012%) and D86G+E91G (0.003%). One isolate had GyrA A92P+ParC D86N alterations, but remained susceptible to gepotidacin (MIC=0.125 mg/L). No GyrA plus ParC alterations resulted in a gepotidacin MIC>4 mg/L. Modelling of gepotidacin binding to GyrA A92/A92T/A92P suggested that gepotidacin resistance due to GyrA A92T might be linked to the formation of a new polar contact with DNA. In silico mining of 33213 global gonococcal genomes (isolates from 1928 to 2023) showed that A92 is highly conserved in GyrA, while alterations in D86 of ParC are common. No GyrA plus ParC alterations caused gepotidacin resistance. MIC determination and genomic surveillance of potential antimicrobial resistance determinants are imperative.

Novel Antibiotics for Urinary Tract Infection (UTI): The EAGLE 2 and EAGLE-3 Trials for Uncomplicated UTI, and the CERTAIN-1 Trial for Complicated UTI.

PMID: 39112136 | Indexed in PubMed

Two novel antibiotics have been evaluated for the treatment of urinary tract infection (UTI). Gepotidacin was an efficacious first-in-class oral antibiotic in uncomplicated UTI in comparison to nitrofurantoin. Cefepime/taniborbactam was superior to meropenem in complicated UTI and acute pyelonephritis.

The Urinary Microbiome: Improving Diagnostics and Management of Urinary Tract Infections in Adult Females.

PMID: 38729665 | Indexed in PubMed

This article discusses the urinary microbiome in relation to urinary tract infection (UTI) in women. It makes biologic sense that the microbiota of different niches (bladder, vagina, and gut) interact with each other in health, as well as during a UTI event; however, these relationships remain poorly understood. Future research should close knowledge gaps regarding the interactions between the urinary microbiota

and the host, amongst the microbiota of adjacent niches, and between the microbes within the same microbiota. The new knowledge should result in improved UTI treatment in the age of antibiotic stewardship.

In women with uncomplicated UTIs, gepotidacin was noninferior to nitrofurantoin for therapeutic response at 10 to 13 d.

PMID: 38830223 | Indexed in PubMed

Wagenlehner F, Perry CR, Hooton TM, et al. Oral gepotidacin versus nitrofurantoin in patients with uncomplicated urinary tract infection (EAGLE-2 and EAGLE-3): two randomised, controlled, double-blind, double-dummy, phase 3, non-inferiority trials. *Lancet*. 2024;403:741-755. 38342126.

Recent dynamics in *Neisseria gonorrhoeae* genomic epidemiology in Brazil: antimicrobial resistance and genomic lineages in 2017-20 compared to 2015-16.

PMID: 38517452 | Indexed in PubMed

Regular quality-assured WGS with antimicrobial resistance (AMR) and epidemiological data of patients is imperative to elucidate the shifting gonorrhoea epidemiology, nationally and internationally. We describe the dynamics of the gonococcal population in 11 cities in Brazil between 2017 and 2020 and elucidate emerging and disappearing gonococcal lineages associated with AMR, compare to Brazilian WGS and AMR data from 2015 to 2016, and explain recent changes in gonococcal AMR and gonorrhoea epidemiology. WGS was performed using Illumina NextSeq 550 and genomes of 623 gonococcal isolates were used for downstream analysis. Molecular typing and AMR determinants were obtained and links between genomic lineages and AMR (determined by agar dilution/Etest) examined. Azithromycin resistance (15.6%, 97/623) had substantially increased and was mainly explained by clonal expansions of strains with 23S rRNA C2611T (mostly NG-STAR CC124) and mtr mosaics (mostly NG-STAR CC63, MLST ST9363). Resistance to ceftriaxone and cefixime remained at the same levels as in 2015-16, i.e. at 0% and 0.2% (1/623), respectively. Regarding novel gonorrhoea treatments, no known zoliflodacin-resistance gyrB mutations or gepotidacin-resistance gyrA mutations were found. Genomic lineages and sublineages showed a phylogenomic shift from sublineage A5 to sublineages A1-A4, while isolates within lineage B remained diverse in Brazil. Azithromycin resistance, mainly caused by 23S rRNA C2611T and mtrD mosaics/semi-mosaics, had substantially increased in Brazil. This mostly low-level azithromycin resistance may threaten the recommended ceftriaxone-azithromycin therapy, but the lack of ceftriaxone resistance is encouraging. Enhanced gonococcal AMR surveillance, including WGS, is imperative in Brazil and other Latin American and Caribbean countries.

Gyrase and Topoisomerase IV: Recycling Old Targets for New Antibacterials to Combat Fluoroquinolone Resistance.

PMID: 38564341 | Indexed in PubMed

Beyond their requisite functions in many critical DNA processes, the bacterial type II topoisomerases, gyrase and topoisomerase IV, are the targets of fluoroquinolone antibacterials. These drugs act by stabilizing gyrase/topoisomerase IV-generated DNA strand breaks and by robbing the cell of the catalytic activities of these essential enzymes. Since their clinical approval in the mid-1980s, fluoroquinolones have been used to treat a broad spectrum of infectious diseases and are listed among the five "highest priority" critically important antimicrobial classes by the World Health Organization. Unfortunately, the widespread use of fluoroquinolones has been accompanied by a rise in target-mediated resistance caused by specific mutations in gyrase and topoisomerase IV, which has curtailed the medical efficacy of this drug class. As a result, efforts are underway to identify novel antibacterials that target the bacterial type II topoisomerases. Several new classes of gyrase/topoisomerase IV-targeted antibacterials have emerged, including novel bacterial topoisomerase inhibitors, Mycobacterium tuberculosis gyrase inhibitors, triazaacenaphthylenes, spiropyrimidinetriones, and thiophenes. Phase III clinical trials that utilized two members of these classes, gepotidacin (triazacenaphthylene) and zoliflodacin (spiropyrimidinetrione), have been completed with positive outcomes, underscoring the potential of these compounds to become the first new classes of antibacterials introduced into the clinic in decades. Because gyrase and topoisomerase IV are validated targets for established and emerging antibacterials, this review will describe the catalytic mechanism and cellular activities of the bacterial type II topoisomerases, their interactions with fluoroquinolones, the mechanism of target-mediated fluoroquinolone resistance, and the actions of novel antibacterials against wild-type and fluoroquinolone-resistant gyrase and topoisomerase IV.

Management of *Neisseria gonorrhoeae* infection: from drug resistance to drug repurposing.

PMID: 38856987 | Indexed in PubMed

Neisseria gonorrhoeae is a common sexually transmitted disease connected with extensive drug resistance to many antibiotics. Presently, only expanded spectrum cephalosporins (ceftriaxone and cefixime) and azithromycin remain useful for its management. New chemotypes for the classical antibiotic drug target gyrase/topoisomerase IV afforded inhibitors with potent binding to these enzymes, with an inhibition mechanism distinct from that of fluoroquinolones, and thus less prone to mutations. The α -carbonic anhydrase from the genome of this bacterium (NgCA α) was also validated as an antibacterial target. By exploiting different subunits from the gyrase/topoisomerase IV as well as new chemotypes, two new antibiotics reached Phase II/III clinical trials, zoliflodacin and gepotidacin. They possess a novel inhibition mechanism, binding in distinct parts of the enzyme compared to the fluoroquinolones. Other chemotypes with inhibitory activity in these enzymes were also reported. NgCA α inhibitors belonging to a variety of classes were obtained, with several sulfonamides showing MIC values in the range of 0.25-4 μ g/mL and significant activity in animal models of this infection. Acetazolamide and similar CA inhibitors might thus be repurposed as anti-infectives. The scientific/patent literature has been searched for on PubMed, ScienceDirect, Espacenet, and PatentGuru, from 2016 to 2024.

Intermethod comparability analyses of gepotidacin antimicrobial susceptibility tests using a large collection of globally collected *Escherichia coli* and *Staphylococcus saprophyticus* clinical isolates.

PMID: 38215519 | Indexed in PubMed

Gepotidacin (GSK2140944) is a novel, bactericidal, first in class triazaacenaphthylene bacterial type II topoisomerase inhibitor in development for the treatment of uncomplicated urinary tract infections and gonorrhea. The performance of several antimicrobial susceptibility methods (broth microdilution, gradient diffusion, and disk diffusion) for gepotidacin were evaluated using over 5800 recent *Escherichia coli* and *Staphylococcus saprophyticus* clinical isolates. Reference broth microdilution gepotidacin MICs showed an essential agreement of 95.9 % and 98.1 % with MICs by gradient diffusion for *E. coli* and *S. saprophyticus* isolates, respectively. Gepotidacin susceptibility using disks produced by 2 manufacturers had good agreement with an R^2 values of 0.95 and 99.2 % of overall zone diameters agreeing within 3 mm. A correlation with an overall R^2 value of 0.72 between MICs by broth microdilution and zone diameters by disk diffusion was observed. This data should assist in the clinical development of gepotidacin and provide reliable susceptibility methods to evaluate its activity.

Gepotidacin: a novel, oral, 'first-in-class' triazaacenaphthylene antibiotic for the treatment of uncomplicated urinary tract infections and urogenital gonorrhoea.

PMID: 36883591 | Indexed in PubMed

The ongoing spread of antimicrobial resistance has made the treatment of uncomplicated urinary tract infections (UTIs) and urogenital gonorrhoea increasingly difficult. New oral treatment options are urgently needed. Gepotidacin (previously GSK2140944) is a novel, bactericidal, oral, 'first-in-class' triazaacenaphthylene antibiotic that inhibits bacterial DNA replication by blocking two essential topoisomerase enzymes. Mutations in both enzymes would likely be necessary for resistance to occur, thus raising hopes that the drug will be able to maintain long-term effectiveness. Data from Phase II clinical trials of gepotidacin in UTIs and urogenital gonorrhoea appear promising, and Phase III trials are underway. In this review we summarize the development of gepotidacin and discuss its potential role in clinical practice. If approved, gepotidacin will be the first new oral antibiotic for UTIs in more than 20 years.